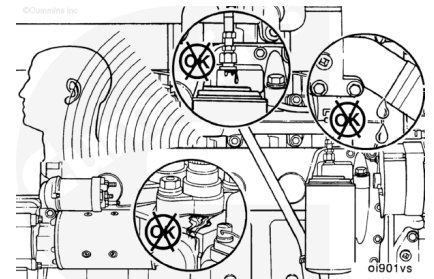


Trainings ▾

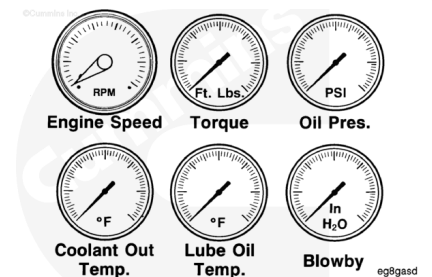
Run-In Instructions

The engine run-in period allows the tester to detect assembly errors and to make final adjustments needed for performance that meets specifications.

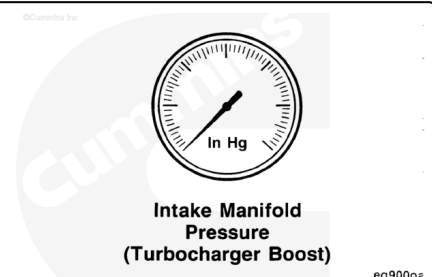
Note : The amount of time specified for the following engine run-in phases are minimums. Additional time can be used, if desired, at each phase except engine idle periods.



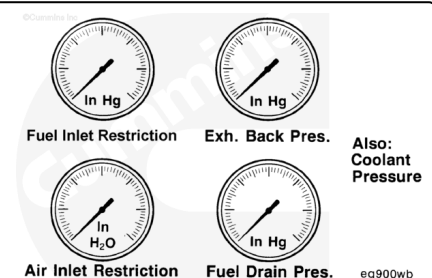
Measurements from these indicators and gauges **must** be observed closely during all phases of the engine run-in period.



To correctly evaluate the engine performance, this additional measurement **must** be observed during engine run-in phases.

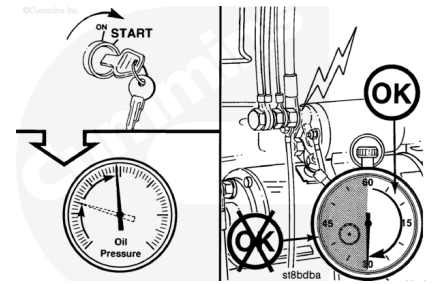


It is good practice to observe these measurements even if engine performance meets specifications. If engine performance does **not** meet specifications, these measurements can indicate possible reasons for under performance.



⚠ CAUTION ⚠

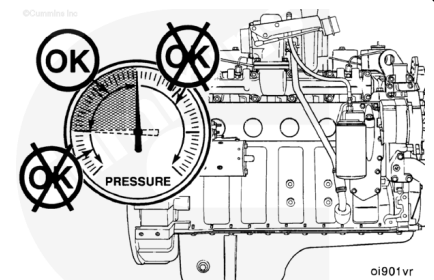
Do not crank the engine for more than 30 seconds. Excessive heat will damage the starting motor.



Crank the engine and observe the lubricating oil pressure when the engine starts. If the engine fails to start within 30 seconds, allow the starting motor to cool for 2 minutes before ranking the engine again.

⚠ CAUTION ⚠

If the lubricating oil pressure is not within specifications, shut off the engine immediately. Low lubricating oil pressure will cause engine damage.

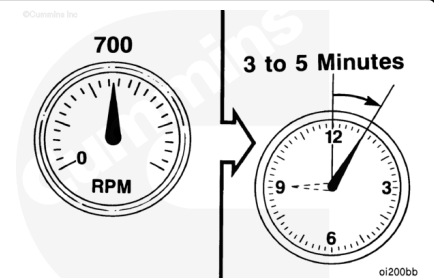


Engine lubricating oil pressure **must** be at least 70 kPa [10 psi] at 700 rpm.

Correct the problem if the lubricating oil pressure is **not** within specifications.

⚠ CAUTION ⚠

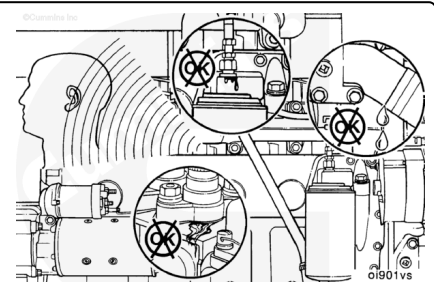
Do not operate the engine at idle speed longer than specified during engine run-in. Excessive carbon formation will cause damage to the engine.



Operate the engine at approximately 700 rpm for 3 to 5 minutes.

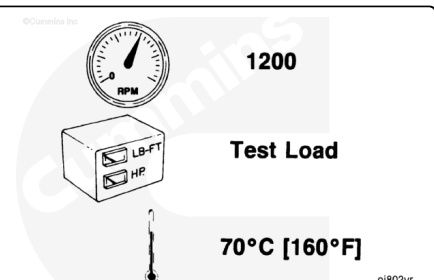
Listen for unusual noises; watch for coolant, fuel, and lubricating oil leaks; and check for correct engine operation in general.

Repair all leaks or component problems before continuing the engine run-in.



Move the throttle to obtain 1,200 rpm engine speed, and set the test load to 25 percent of the rated load.

Operate the engine at this speed and load level until the coolant temperature is 70° C [160° F]. Check all gauges and record the data.

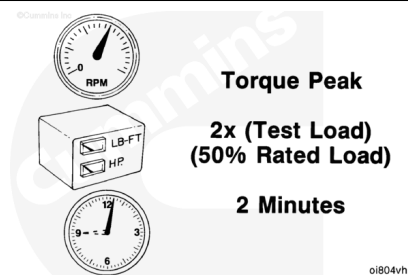


Do **not** proceed to the next step until a steady blowby reading is obtained.

Open the throttle to the speed at which peak torque occurs, and adjust the dynamometer load to 50 percent of torque peak load. Operate the engine at this speed and load level for 2 minutes.

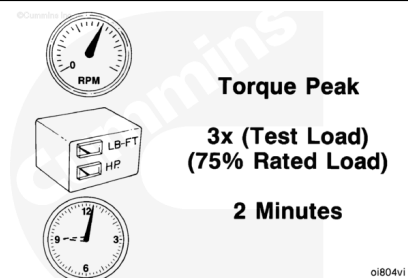
Check all gauges and record the data.

Do **not** proceed to the next step until blowby is stable within specifications.



With the engine speed remaining at torque peak rpm, increase the dynamometer load to 75 percent of torque peak load. Operate the engine at this speed and load level for 2 minutes. Check all gauges and record the data.

Do **not** proceed to the next step until blowby is stable within specifications.



Move the throttle lever to its fully opened position, and increase the dynamometer load until the engine speed is at torque peak rpm. Operate the engine at this speed and load level for 10 minutes or until the blowby becomes stable within specifications.

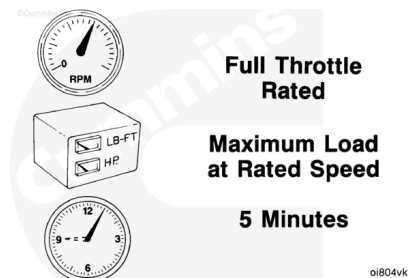
Check all gauges and record the data.



Reduce the dynamometer load until the engine speed increases to the engine's rated rpm.

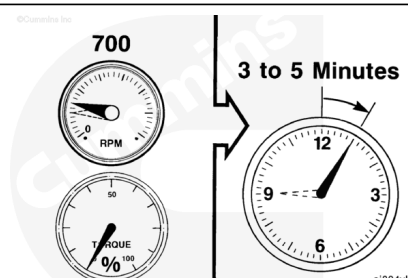
Operate the engine at rated rpm for 5 minutes.

Check all gauges and record the data.



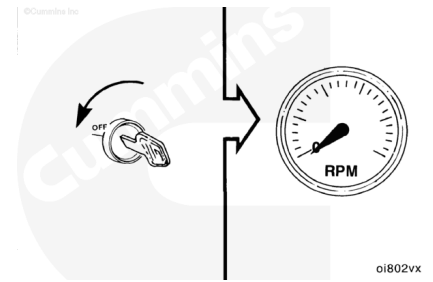
⚠ CAUTION ⚠

Shutting off the engine immediately after operating at full load will damage the turbocharger and internal components. Always allow the engine to cool before shutting it off.



Remove the dynamometer load completely, and operate the engine at 700 rpm for 3 to 5 minutes. This period will allow the turbocharger and other components to cool.

Shut off the engine.

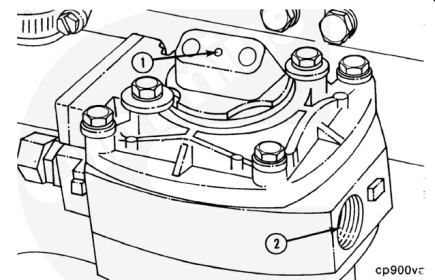


Engine Dynamometer Test - Performance Checking

Make sure the air compressor is unloaded during the performance check.

Apply regulated air pressure of 655 kPa [95 psi] to the air compressor unloader (1).

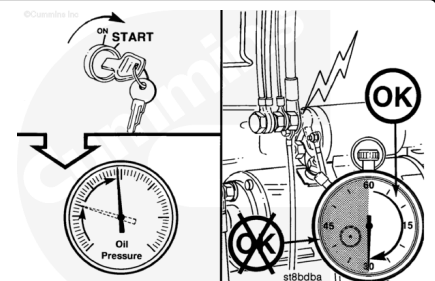
The compressed air load in the accompanying illustration **must** be attached to the air compressor outlet (2).



⚠ CAUTION ⚠

Do not crank the engine for more than 30 seconds. Excessive heat will damage the starting motor.

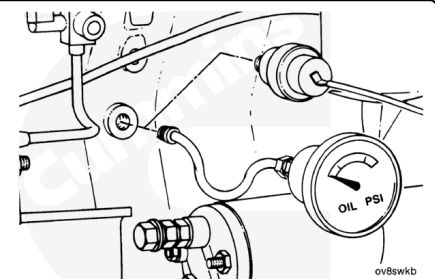
Crank the engine and observe the oil pressure when the engine starts. If the engine fails to start within 30 seconds, allow the starting motor to cool for 2 minutes before cranking the engine again.



⚠ CAUTION ⚠

If the lubricating oil pressure is not within specifications, shut off the engine immediately. Low lubricating oil pressure will cause engine damage. Correct the problem if lubricating oil pressure is not within specifications.

Engine lubricating oil pressure **must** be at least 69 kPa [10 psi] at approximately 700 rpm.



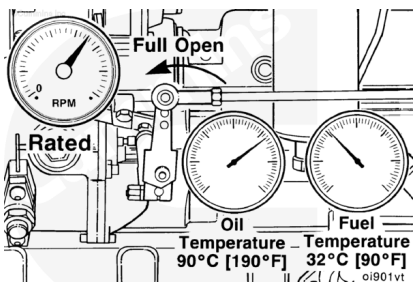
Make sure the engine is at operating temperature.

Move the throttle lever to the "FULL OPEN" position. Adjust the dynamometer load until the engine maintains the rated rpm.

Allow the readings to stabilize. Read the horsepower.

Check all the gauges, and record the readings.

The horsepower reading will **not** be accurate if the lubricating oil temperature and fuel temperature are **not** within specifications.



Lubricating Oil Temperature		
celsius		fahrenheit
90	MAX	190

Fuel Temperature		
celsius		fahrenheit
32	MAX	90

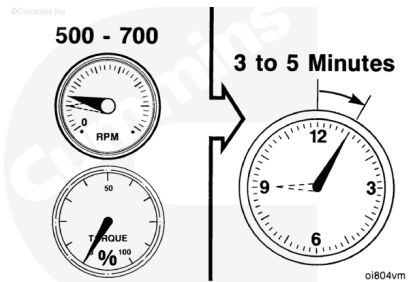
Check all gauges and record the data.

⚠ CAUTION ⚠

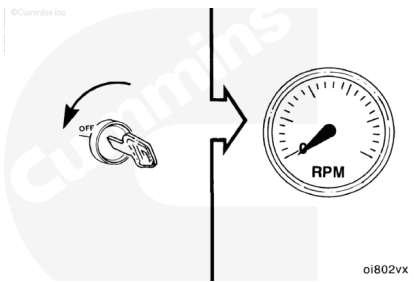
Do not shut off the engine immediately after it has been loaded. It must be allowed to sufficiently cool.

Remove the dynamometer load completely, and operate the engine at idle speed for 3 to 5 minutes. This will allow the turbocharger and other components to cool.

Idle periods longer than 5 minutes are to be avoided.



Shut off the engine after the cool down period.



Remove all test instrumentation. Remove the engine from the dynamometer.

If the engine is to be stored temporarily and does **not** have permanent type antifreeze, it is necessary to drain all coolant.

